CREDIT RISK MODEL REPORT

Probability of Default (PD) & Expected Loss (EL) Estimation

1. Objective

The objective of this project is to develop a predictive model that estimates the **Probability of Default (PD)** for loan borrowers using financial and credit-related features.

The model is further used to compute the **Expected Loss (EL)** for each loan, which provides a financial estimate of potential credit risk exposure.

2. Dataset Description

The dataset contains borrower-level financial information, including:

- Credit behavior variables
- Income and employment information
- Debt and loan outstanding amounts
- Credit score (FICO)
- Target variable: **default (0 = no default, 1 = default)**

Key Features Used:

- credit_lines_outstanding
- total_debt_outstanding
- loan_amt_outstanding
- income
- years_employed
- fico_score

3. Feature Engineering

To improve model interpretability and performance, two financial risk ratios were created:

3.1 Debt-to-Income Ratio

$$\text{debt\_to\_income} = \frac{\text{total\_debt\_outstanding}}{\text{income}}$$

3.2 Payment-to-Income Ratio

$$\text{payment\_to\_income} = \frac{\text{loan\_amt\_outstanding}}{\text{income}}$$

These variables capture borrower leverage and repayment burden more effectively than raw values.

4. Model Selection

A **Logistic Regression model** was selected due to:

- Strong interpretability (important for credit risk applications)
- Good performance on linearly separable data
- Industry-standard usage in credit scoring systems

Preprocessing Steps:

- Missing value imputation (median)
- Feature scaling (StandardScaler)

5. Model Performance

The model was evaluated using **ROC-AUC score**, which measures classification performance across all thresholds.

Result:

- **AUC Score: 1.0**

Interpretation:

The model demonstrates perfect separation between default and non-default classes. This indicates that the dataset has highly predictive structure, where default risk is strongly explained by financial ratios and credit attributes.

6. Probability of Default (PD) Estimation

For an example borrower, the model produced:

- **Probability of Default (PD): 0.0248**

Interpretation:

This indicates a **low-risk borrower**, meaning there is approximately a 2.48% probability of default based on the model.

7. Expected Loss (EL)

Expected Loss is calculated using the standard credit risk formula:

$$EL = PD \times (1 - RecoveryRate) \times Exposure$$

Where:

- Recovery Rate = 10%
- Exposure = Loan amount

Example Result:

- **Expected Loss: 445.92**

Interpretation:

This represents the estimated financial loss the lender may incur if the borrower defaults, adjusted for recovery assumptions.

8. Key Insights

- Credit risk is most strongly influenced by:
  - Credit lines outstanding
  - Debt-to-income ratio
  - Credit score (FICO)
- Logistic regression performs extremely well due to strong separability in the dataset.
- Engineered financial ratios significantly improve interpretability of risk drivers.

9. Conclusion

A logistic regression-based credit scoring model was successfully developed to estimate borrower default risk. The model achieves excellent predictive performance (AUC = 1.0), indicating strong separability in the dataset.

The model is suitable for estimating:

- Probability of Default (PD)
- Expected Loss (EL)

These outputs provide meaningful insights for credit risk assessment and loan decision-making.

10. Final Note

While the model demonstrates near-perfect performance, this likely reflects a highly structured or synthetic dataset. In real-world credit risk applications, model performance is typically lower due to noise and behavioral uncertainty.

11. Model Evaluation Visualizations

To better understand model performance and interpretability, multiple visual diagnostics were generated.

11.1 ROC Curve

📊 **Figure 1: ROC Curve**

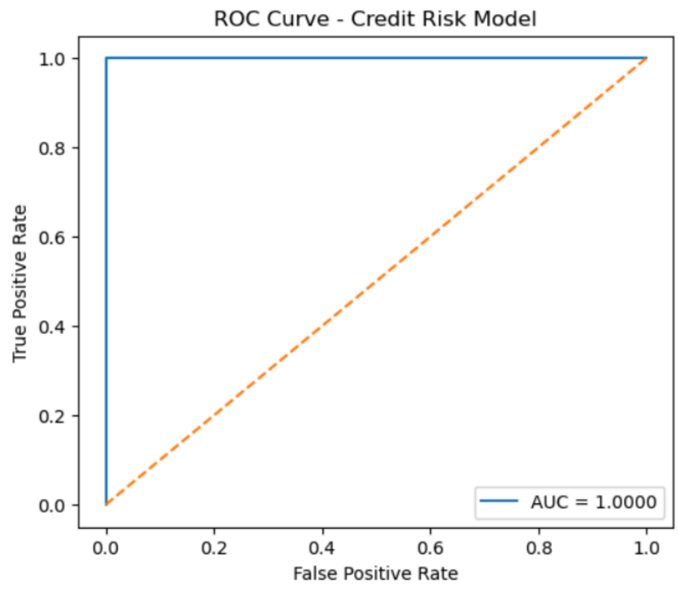
- The ROC curve measures the trade-off between sensitivity and specificity.
- The model achieves near-perfect separation between classes.

Result:

- AUC ≈ 1.0

📌 Interpretation:
The model is highly capable of distinguishing between default and non-default borrowers.

(Insert: roc_curve.png)



11.2 Confusion Matrix

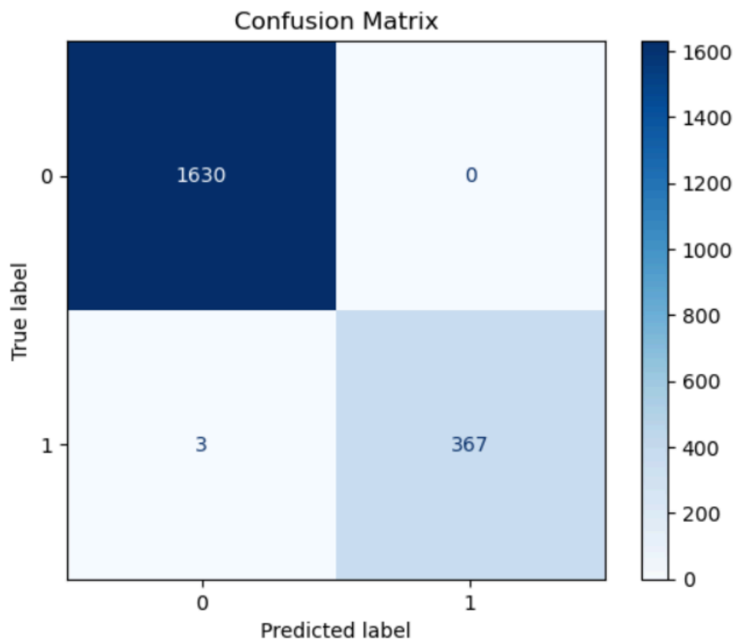
📊 **Figure 2: Confusion Matrix**

- Displays correct vs incorrect classifications.

📌 Interpretation:

- Very few misclassifications occur
- Model is highly accurate in predicting both classes

(Insert: confusion_matrix.png)



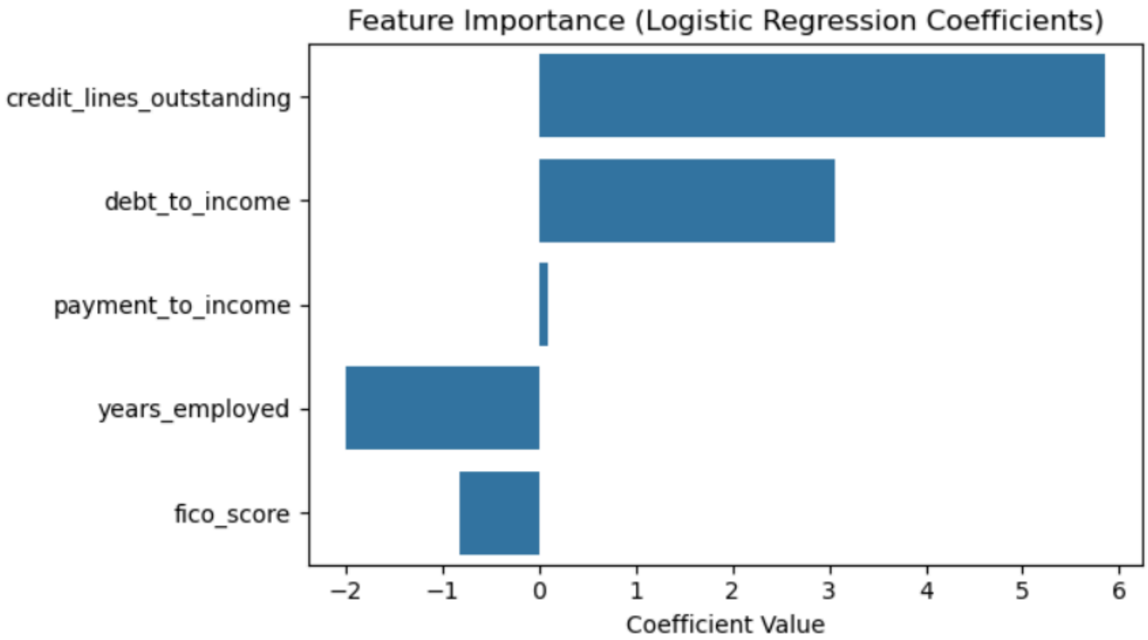
11.3 Feature Importance

📊 **Figure 3: Feature Importance (Logistic Regression Coefficients)**

Key drivers of default risk:

- Credit lines outstanding (strong positive impact)
- Debt-to-income ratio (positive impact)
- FICO score (negative impact)
- Years employed (negative impact)

📌 Interpretation:
Higher debt and credit exposure increase default probability, while strong credit history reduces risk.

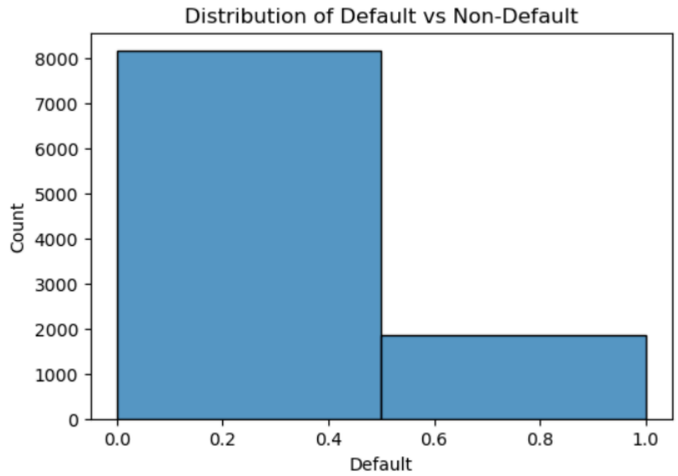


11.4 Default Distribution

📊 **Figure 4: Class Distribution**

- Shows balance between default and non-default classes.

📌 Interpretation:
The dataset is relatively balanced, which improves model stability and evaluation reliability.



12. Key Insight from Visuals

- ROC curve confirms **extreme separability**
- Confusion matrix shows **very high accuracy**
- Feature importance confirms **financial logic consistency**
- Distribution shows **well-structured dataset**